

CS471 – Lesson 20 Solutions Key

Probabilistic Systems: Bayesian Networks

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Bayesian Network Structure & Family Relationships

Scenario: Consider the tactical air-strike planning network from Quiz V:

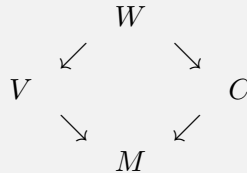
$$W \rightarrow V, \quad W \rightarrow C, \quad V \rightarrow M, \quad C \rightarrow M$$

where W is Weather, V is Optical Visibility, C is SATCOM Communications, and M is Mission Execution.

- a. **Question:** Sketch or diagram the Directed Acyclic Graph (DAG) for this network.

Solution

W has directed edges pointing down to V and C . Both V and C have directed edges pointing down into M .



- b. **Question:** List the immediate **parents** of M :

Solution

V and C. (These are the only nodes with direct incoming edges pointing into M).

- c. **Question:** Is W an **ancestor** of M ? Circle one: **YES** / **NO**

Solution

YES. An ancestor is any node from which a directed path leads to M . Here, directed paths exist: $W \rightarrow V \rightarrow M$ and $W \rightarrow C \rightarrow M$.

- d. **Question:** Complete the conditional-independence statement: $V \perp C \mid \underline{\hspace{2cm}}$
Explain why observing this variable renders V and C conditionally independent.

Solution

W ($V \perp C \mid W$).

Explanation: W is the common cause of both V and C (diverging fork). When W is unobserved, knowing V gives evidence about W , which informs C . Once W is known/conditioned on, the information flow between V and C is blocked.

2. Joint Distribution Factorization

- a. **Question:** Write the mathematical chain rule factorization for the joint distribution $P(W, V, C, M)$ using the parents of each node.

Solution

$$P(W, V, C, M) = P(W) \cdot P(V \mid W) \cdot P(C \mid W) \cdot P(M \mid V, C)$$

- b. **Question:** Calculate the total number of independent probability parameters required to populate this Bayesian network versus an unconstrained full joint probability table.

Solution

Unconstrained Table: For 4 binary variables, $2^4 - 1 = \mathbf{15}$ parameters.

Bayesian Network CPTs:

- $P(W)$: $2^0 = 1$ parameter
- $P(V \mid W)$: $2^1 = 2$ parameters
- $P(C \mid W)$: $2^1 = 2$ parameters
- $P(M \mid V, C)$: $2^2 = 4$ parameters

Total = $1 + 2 + 2 + 4 = \mathbf{9}$ parameters.

Part 2: Student Independent Practice Solutions

A. Autonomous Air Defense Network

Scenario: Edges are $R \rightarrow D$, $T \rightarrow D$, $T \rightarrow I$, $D \rightarrow F$, $I \rightarrow F$.

1. **Question:** List all immediate **parents** of D .

Solution

$$\text{Parents}(D) = \{\mathbf{R}, \mathbf{T}\}.$$

2. **Question:** List all **descendants** of T .

Solution

$$\text{Descendants}(T) = \{\mathbf{D}, \mathbf{I}, \mathbf{F}\}. \text{ (All nodes reachable downstream from } T\text{).}$$

3. **Question:** Complete the conditional independence statement: $D \perp I \mid \underline{\hspace{2cm}}$.

Solution

$$D \perp I \mid \mathbf{T} \text{ (or } \{T, R\}\text{).}$$

Terrain Clutter (T) is the common cause parent connecting D and I .

4. **Question:** Write the full joint probability decomposition $P(R, T, D, I, F)$ in terms of local conditional distributions.

Solution

$$P(R, T, D, I, F) = P(R) \cdot P(T) \cdot P(D \mid R, T) \cdot P(I \mid T) \cdot P(F \mid D, I)$$

B. Joint Probability Computation

Scenario: Compute $P(R = 1, T = 1, D = 1, I = 1, F = 1)$.

Solution

Step 1: Apply the factored equation:

$$P(R = 1, T = 1, D = 1, I = 1, F = 1) = P(R = 1)P(T = 1)P(D = 1 \mid R = 1, T = 1)P(I = 1 \mid T = 1)P(F = 1 \mid R = 1, T = 1, D = 1, I = 1)$$

Step 2: Substitute values:

$$= (0.30) \times (0.40) \times (0.10) \times (0.70) \times (0.90)$$

Step 3: Multiply sequentially:

$$0.30 \times 0.40 = 0.12$$

$$0.12 \times 0.10 = 0.012$$

$$0.012 \times 0.70 = 0.0084$$

$$0.0084 \times 0.90 = \mathbf{0.00756} \quad (0.756\%)$$